1.4.5 Misconceptions

A major misconception about Elasticsearch is mistaking it for a traditional relational database. It's also a common misunderstanding that setting up Elasticsearch is easy, while in reality, many tweaks are required to set up a decent-sized cluster. In addition, Elasticsearch is often thought of as a technology used for text search, when in fact it can be used for a wide range of search and analytics use cases. The following list summarizes some common misconceptions about Elasticsearch:

- *Elasticsearch is easy to set up and manage.* While Elasticsearch is relatively straightforward to set up and get started with, it can be challenging to manage and scale as data grows and use cases increase. Although everything works out of the box, making an engineer's life easy, taking Elasticsearch into the production environment requires effort. We may need to tweak the configuration and fine-tune the memory, manage node failures, or even scale the cluster to handle petabytes of data as our data grows.
- *Elasticsearch is a relational database.* Elasticsearch is *not* a relational database and does not support traditional relational database features like transactions, foreign keys, and complex join operations. For example, we can't enforce referential integrity or perform complex join operations in Elasticsearch. If you need these features, a proven relational database like MySQL or PostgreSQL is definitely your solution.
- *Elasticsearch can handle all types of data.* Elasticsearch is versatile and can handle a wide range of data types, but it is not designed to handle every type of data with equal ease. For example, it may not be the best solution for real-time data processing and analytics or for handling large binary data. If you need to store and process large binary data like videos or images, consider using a dedicated binary data store like HDFS or Amazon S3.
- *Elasticsearch is only for text search.* While Elasticsearch is great for text search, it can also perform complex analytics on structured and unstructured data. For example, we can use Elasticsearch to perform aggregations, analyze log data, and visualize data using Kibana.
- *Elasticsearch can replace all other technologies.* Elasticsearch is a powerful and flexible technology, but it is not a one-size-fits-all solution and or the best choice for every use case. It can never replace a traditional relational database, for example.
- *Elasticsearch is always faster than other technologies.* Elasticsearch is designed for high performance and is expected to perform well under heavy loads. However, there's only so much Elasticsearch can do, and its performance primarily depends on how well the platform engineers fine-tune it.
- *Elasticsearch deals with big data only.* Elasticsearch can handle petabytes of data in large data sets, but it is equally performant when dealing with small data sets on the order of a few gigabytes. For example, we can use Elasticsearch to search and analyze data for an organization's small email database or a startup company without much effort.